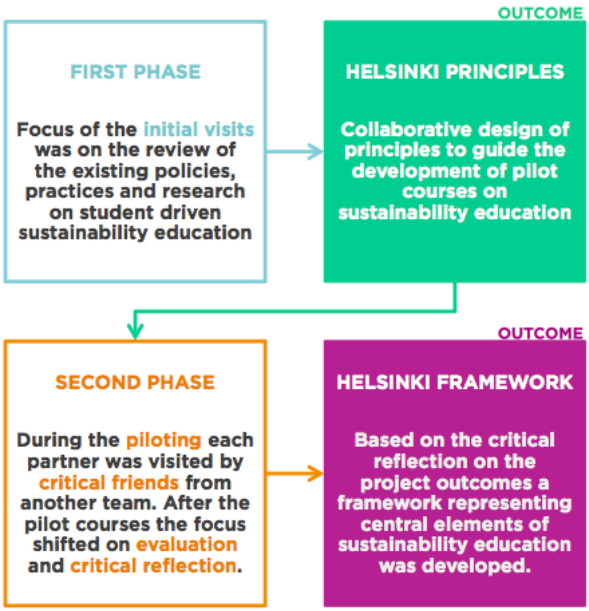
HELSINKI FRAMEWORK:
CO-DESIGN OF PRINCIPLES FOR SUSTAINABILITY EDUCATION

Veli-Matti Vesterinen
University of Turku

In this chapter I discuss the creation of the Helsinki Framework, which is one of the major outcomes of ActSHEN. It represents elements that project team members saw as central to working with sustainability education. The purpose of the framework is to provide ideas for embedding sustainability awareness and action in higher education. The creation of the framework was a messy affair, as such collaborative design processes tend to be. To make sense of this mess, I will present here a design narrative (see Bell, Hoadley & Linn, 2004), which describes what happened during the design process. It is written from my personal point of view.

DEVELOPMENT OF THE HELSINKI FRAMEWORK

The development of the Helsinki Framework was carried out in two phases. At the end of the first year, a set of guiding principles was defined in a participatory design process that sought to capture, coalesce and synthesize previous research, relevant policies and examples of good practice on student-driven education for sustainability in higher education. In the second phase, these guiding principles were employed in the design, development, and assessment of education initiatives in each of the institutions involved in the project. The iterative development of the guiding principles finally culminated in creation of a framework of four core elements of sustainability education.



Development of the Helsinki Principles and the Helsinki Framework

FIRST PHASE

During the first phase, the project partners visited partner institutions to familiarize themselves with each other's current practices on sustainability education and to discuss relevant research and policies. The first meeting began with a review of policies on sustainability and student-driven initiatives in each Nordic country and partner universities. Through exchange visits, we learned about the various models, contributors and constraints of student-driven education for sustainability.

As the partner and partner institutions had a wide variety of settings and goals, such collaborative knowledge building played a crucial part in creating a community of practitioners capable of finding agreement on the core elements of student-driven education for sustainability on higher education.

At the end of the first phase, we decided that developing a single model representing student-driven education for sustainability could be limiting and restrictive. Thus we decided to focus on development of a set of more general guiding principles.

In the development of the principles we used our own experience as well as the insights gained in the workshops during the first year of the project. One of the key inputs were the existing student-driven education models, such as the student coordinator model used by CEMUS (see Hald 2011) and collaborative peer teaching models developed utilising educational design research methodology at the Unit of Chemistry Teacher Education at the University of Helsinki (see Vesterinen & Aksela, 2013).

Whilst the original principles set out in the project proposal suggested an implied understanding of the characteristics of sustainability in higher education, the Helsinki meeting in May 2014 demonstrated a need to develop these into a concrete set of working principles drawn from their diverse understandings, knowledge, experiences, and expectations of the pilot initiatives. Although there was no agreement as regards recognizing these principles as a final representation of sustainability education, there was consensus that the Helsinki principles, as they are referred to in this document, should guide the development, implementation, and assessment of the pilot courses and initiatives.

SECOND PHASE

During the second phase, the principles were used by the participating institutions as a framework to guide the development, implementation and assessment of pilot courses or initiatives. The focus on a set of general guiding principles instead of one-size-fits-all pedagogic models provided a common starting point for project partners, whilst allowing for the local customization of the courses.

An important element of the work during this phase was the critical friend visits. The concept of the critical friend can be defined in the following way:

A critical friend, as the name suggests, is a trusted person who asks provocative questions, provides data to be examined through another lens, and offers critiques of

a person's work as a friend. A critical friend takes the time to fully understand the context of the work presented and the outcomes that the person or group is working toward. The friend is an advocate for the success of that work. (Costa & Kallick, 1993, p. 50)

Working with this understanding, each partner was visited by critical friends from another team. The results of the critical friend visits were used by the respective institutions to make modifications to and development of pilot initiatives, and in some cases, future courses. Inevitably, the process and outcome of the visits impacted on and influenced the participants on an individual level, generating self-reflection on the nature of and interplay between pedagogical practices, professional roles, and institutional challenges.

During the end of the second phase, the focus of the project shifted towards evaluation of the progress made in the pilot initiatives, as well as the iterative refinement of the principles used to guide the pilot initiatives. Project partners also reflected on the project process and their role within it in relation to others. Although course evaluation was carried out by course developers, it also involved critical friends and continuous self-reflections.

An important part of the self-development process was the collaboration amongst project members when we met for conference presentations and project meetings. These small gatherings provided an opportunity to formally and informally work with ideas and experiences that came out of the pilot initiatives. The development of these emerging new ideas led us to review and refine the Helsinki principles as part of the evaluation of the project. The refinement of the Helsinki principles towards a final framework was as an integral part of the project in that it reflects and represents the learning process that took place.

In March 2016, after two years of experimenting, the project team met to conduct a final critical reflection on the project outcomes and how these should be documented. The meeting was facilitated by Stephen Sterling acting as a critical friend (see Costa & Kallick, 1993). It was during this meeting that the Helsinki principles were revisited and reviewed in terms of their purpose now that the pilot initiatives had come to an end. The outcome of the critical dialogue was that the Helsinki principles should be developed into a final framework, referred to as the Helsinki Framework, representing elements that project team members believe to be central to working with sustainability education.

REFLECTION

The creation of the Helsinki principles was one of the integral framing actions during the ActSHEN project. In previous research on collaborative design (e.g. Hey, Joyce & Beckman 2007; Ylirisku et al. 2009) framing has been defined as the process by which people consciously or unconsciously structure the situation to make the decisions. Frame provides structure from the viewpoint of an actor, highlights as well as hides different elements and includes assumptions of a desired end state. As the

partners had a wide variety of settings and goals, collaborative knowledge building played a crucial part in negotiating a common framework.

Key framing activities included writing of the project proposal, biannual meetings at partner institutions, devising the first draft of the framework to inform the pilot projects and courses, visits from critical friends, preparing of conference papers and presentations, revision of the framework facilitated by an expert, as well as writing an online report detailing the project and most of the pilot projects. These activities enabled us to capture, coalesce and synthesize previous research, relevant policies and examples of good practice to create a cohesive framework of the core elements of student--driven education for sustainability on higher education. The diversity in the ways to collaborate enabled us to form widely shared frames without the forcing us to reach cognitive consensus. From my own personal viewpoint this allowed the pluralism of viewpoints, which supports creation of more creative solutions than a group with a forced frame of reference.

REFERENCES

Bell, P., Hoadley, C. M., & Linn, M. C. (2004). Design-based research in education. In M. C. Lynn (ed.), *Internet environments for science education, pp.* 73-85. London: Routledge.

Costa, A. L., & Kallick, B. (1993). Through the lens of a critical friend. *Educational leadership, 51*, 49-49.

Hald, M. (ed.) (2011). *Transcending boundaries: How CEMUS is changing how we teach, meet and learn*. Uppsala, Sweden: Cemus – Uppsala Centre for Sustainable Development, Uppsala University and SLU. http://www.web.cemus.se/wp-content/uploads/2015/05/TranscendingBoundaries.pdf

Hey, J. H., Joyce, C. K., & Beckman, S. L. (2007). Framing innovation: negotiating shared frames during early design phases. *Journal of Design Research, 6*(12), 7999. http://doi.org/10.1504/JDR.2007.015564

Vesterinen, V. M., & Aksela, M. (2013). Design of chemistry teacher education course on nature of science. *Science & Education, 22*(9), 2193-2225. http://doi.org/10.1007/s11191-012-9506-0

Ylirisku, S., Halttunen, V., Nuojua, J., & Juustila, A. (2009, April). Framing design in the third paradigm. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 1131-1140). http://doi.org/10.1145/1518701.1518874